

Sagar Hasan

Software Engineer (Full Stack) | 2+ years experience

BSc in Computer Science & Engineering, 2024

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SUMMARY

Software Engineer with 2+ years of experience building scalable, maintainable enterprise web applications. Skilled in Node.js, Express and OOP design patterns. Strong analytical skills with 1500+ problems solved on [LeetCode](#). Proactive, fast learner, fluent in English.

TECHNICAL SKILLS

- **Languages:** JavaScript (ES6+, async/await, closures, modules), Python, C/C++
- **Backend:** Node.js/Express.js, Socket.io, REST APIs.
- **Frontend:** React (SPA framework), MUI (Material-UI), Redux, Vite, HTML, CSS, JavaScript
- **Database:** MySQL, SQL (queries, JOINS, indexing, transactions), MongoDB
- **Tools:** Git, GitHub, Node.js Inspector, Postman, Chrome DevTools, VS Code Debugger
- **Core Concepts:** DSA OOP, SOLID Principles, Design Patterns (Repository, Factory), System Design, Edge Case Analysis.

EXPERIENCE

Software Engineer - LYXA S.A.L

08/2024 – present | Gulshan 1, Dhaka, Bangladesh

- Built RESTful APIs using Node.js/Express.js + MongoDB, reducing response time by 25%.
- Applied OOP and Repository pattern to refactor legacy code, improving maintainability.
- Designed a **rider assignment system** using graph algorithms, improving delivery efficiency.
- Estimated and delivered 5 features in **order-flow** on schedule with <5% rework.
- Participated in code reviews and knowledge transfer sessions.

Software Engineer Intern - LYXA S.A.L

02/2024 – 07/2024 | Gulshan 1, Dhaka, Bangladesh

- Assisted in debugging and improving system performance through structured issue tracking.
- Identified, reproduced, and resolved critical bugs in Console and backend services.
- Wrote reusable and maintainable code, improving system stability and performance.
- Collaborated with cross-functional teams to improve product quality and user experience.

PROJECTS

Val5th-Finder

[Client-Repo](#) [Server-Repo](#) [Live](#)

Developed a full-stack real-time lobbymaking platform for Valorant players to find teammates ("5th player") based on rank, roles, and playstyle.

- Built a scalable backend using **Node.js, Express, MongoDB, Mongoose, and TypeScript**.
- Implemented Google OAuth 2.0 authentication, replacing traditional JWT-based auth.
- Designed real-time features with **Socket.IO** (Player lobbies, region-based events, live updates).
- Designed a **Lobby distribution algorithm** to match players based on rank, roles, region, and availability.
- Integrated geolocation and IP-based detection to automatically select optimal Valorant regions.
- Developed a responsive frontend using **React, TypeScript, Redux Toolkit, and Material UI**.

Talk2Active – Language Practice Platform & Wisdom feed

[Client-Repo](#) [Server-Repo](#) [Live](#)

- Designed backend architecture using **TypeScript, Node.js, MongoDB**.
- Designed and implemented a **Post distribution algorithm** and **Room-splitting algorithm** to optimize task allocation and improve system scalability.
- Build a responsive frontend using **React, TypeScript, Redux Toolkit, and Material UI**.

File Processing kit - No Data Uploads Required

[Git Repo](#) [Live](#)

- Engineered a zero-server file processing platform with 6 client-side tools (image editor, PDF merge/compress, OCR, batch PDF conversion, resizer) leveraging React hooks, Canvas API, pdf-lib, and Tesseract.js.

TESTING & QUALITY EXPERIENCE

- Wrote test scenarios and manually validated APIs and system flows.
- Identified edge cases and reported bugs with clear documentation.
- Improved system performance by debugging logic and optimizing workflows.
- Strong focus on product quality, reliability, and performance analysis.

EDUCATION

Bachelor of Science: Computer Science and Engineering

Daffodil International University - Savar, Dhaka, Bangladesh

September, 2020 - February, 2024

Relevant coursework: Data Structures, Algorithms, Object-Oriented Programming, Databases, Operating Systems, Computer Networking

English: Professional working proficiency

INTEREST

- System Design & Operating system
- Competitive Programming (analysis & synthesis)
- Product Quality Improvement